

Translation of the exponential comparison: full coefficient, period, metric, and finite-field maps

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1 The original fibre and the auxiliary translation

The September 13 exponential comparison starts with the actual arithmetic polynomial and sum operator

$$\chi(S) = S^q + \sum_{b=0}^{q-1} c_b S^b, \quad q > 0, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S,$$

and the specified primitive

$$\Phi(S) = \frac{S^{q+1}}{q+1} + \sum_{b=0}^{q-1} \frac{c_b S^{b+1}}{b+1}.$$

Its cohomology is the cokernel of $D_{u,t} = u\partial_S + \chi(S) - t$, with the omitted differential symbol understood as the coefficient identification with one-forms. All formulas below retain this original object at $a = 0$. The parameter $a \in \mathbb{C}$ defines an auxiliary comparison family; a translated polynomial is not asserted to be a new zeta zero packet.

Set

$$\chi_a(S) = \chi(S - a), \quad \Phi_a(S) = \Phi(S - a) - \Phi(-a), \quad T_a P(X) = P(X + a). \quad (\text{DT.1})$$

Thus $\Phi_a(0) = 0$, and no additive constant in the potential is lost. Use the same substitution on coefficients of one-forms, with $dS = dX$.

Proposition 1 (The coefficient and connection maps). *The map T_a is an isomorphism of the two-term polynomial complexes for χ_a and χ , over $\mathbb{C}[u, t]$. It induces an isomorphism of their free rank- q cohomology modules. On the special fibre $(u, t) = (0, 0)$,*

$$T_a : E_a = \mathbb{C}[S]/(\chi_a) \xrightarrow{\sim} E, \quad T_a A_a T_a^{-1} = A + aI. \quad (\text{DT.2})$$

Every primary multiplicity and nilpotent Jordan block is retained. For $u \neq 0$, the descended parameter connections satisfy

$$T_a \nabla_t^a T_a^{-1} = \nabla_t - \frac{a}{u} I, \quad \nabla_t = \partial_t - \frac{A(t)}{u}, \quad (\text{DT.3})$$

where $A(t)$ is the original companion matrix for $\chi - t$.

Proof. Substitution gives, for every polynomial P ,

$$T_a(uP' + (\chi_a - t)P) = u\partial_X(T_a P) + (\chi(X) - t)T_a P.$$

The inverse is substitution by $-a$. A polynomial of degree n with leading coefficient $b \neq 0$ is sent by either differential to a polynomial of degree $n + q$ with the same leading coefficient. Successive subtraction of such images therefore gives a unique representative of degree less than q , over the coefficient ring. This proves the stated freeness and the base-change identification. On the special

fibre, $T_a(SP) = (X + a)T_aP$, which proves DT.2. In particular, $(A_a - (\lambda + a)I)^m$ is conjugate to $(A - \lambda I)^m$ for every m , proving the multiplicity statement without discarding the nilpotent part. On arbitrary polynomial representatives, the connection is $\partial_t - S/u$. Since T_a is independent of t , its conjugate is $\partial_t - (X + a)/u$. The commutator with the differential vanishes, so this equality descends and yields DT.3. $\square$

In the fixed bases $1, S, \dots, S^{q-1}$ and $1, X, \dots, X^{q-1}$, the matrix of T_a is

$$(C_a)_{rb} = \begin{cases} \binom{b}{r} a^{b-r}, & 0 \leq r \leq b < q, \\ 0, & r > b, \end{cases} \quad C_a^{-1} = C_{-a}, \quad \det C_a = 1. \quad (\text{DT.4})$$

The inverse and composition rules follow by composing the literal substitutions. In particular, $C_a A_a(t) C_a^{-1} = A(t) + aI$.

2 Periods and the retained exponential factor

Fix $u \neq 0$, put $d = q + 1$, and use the original rays

$$\theta_j = \frac{\arg u + \pi}{d} + \frac{2\pi j}{d}, \quad \Gamma_j = -\ell_0 + \ell_j, \quad 1 \leq j \leq q,$$

oriented as in the exponential comparison. Let $\Pi(u, t)$ denote its period matrix, whose (j, b) entry is $\int_{\Gamma_j} e^{(\Phi(X) - tX)/u} X^b dX$. Let Π_a use the identical labelled ray prescription for Φ_a in the S -plane.

Proposition 2 (Exact period translation). *Define*

$$h_a(t) = -\Phi(-a) - ta, \quad f_a(u, t) = \exp(h_a(t)/u). \quad (\text{DT.5})$$

Then

$$\Pi_a(u, t) = f_a(u, t) \Pi(u, t) C_a. \quad (\text{DT.6})$$

The map $f_a T_a$ intertwines the two parameter connections in DT.3.

Proof. The substitution $S = X + a$ gives the exact identity

$$\Phi_a(S) - tS = \Phi(X) - tX + h_a(t).$$

It changes the contour to $\Gamma_j - a$. This translated contour may be moved to Γ_j without changing the integral. Indeed, truncate the two rays at radius R , join their translated endpoints by segments of length $|a|$, and apply Cauchy's theorem to the resulting closed path. The entire integrand has no finite singularity. On each joining segment its leading exponent is $-R^d/(d|u|) + O(R^{d-1})$, uniformly along that segment; a polynomial factor grows at most polynomially. Its integral tends to zero. The two portions at the finite starting point have opposite orientation and cancel. Absolute convergence on the original and translated rays follows from the same estimate. Finally, $(X + a)^b$ is the column expansion DT.4. This proves DT.6 including the scalar and all phases.

Since $\partial_t f_a = -(a/u)f_a$, for every section v one has

$$\nabla_t(f_a T_a v) = f_a \left(\nabla_t T_a v - \frac{a}{u} T_a v \right) = f_a T_a \nabla_t^a v.$$

The scalar f_a is nonzero on $u \neq 0$; no extension of this exponential gauge to $u = 0$ is used. DT.2 is the separately proved coefficient specialization at that fibre. $\square$

3 The original positive forms and their control operators

The period matrices used below are invertible on the stated domain $u \neq 0$. Here is a complete verification needed for the determinant ratios. At the monomial potential, the substitution $x = r^d/(d|u|)$ in the absolutely convergent ray integral gives

$$\Pi_{j,b}^{\text{mon}} = \frac{(d|u|)^{(b+1)/d}}{d} \Gamma\left(\frac{b+1}{d}\right) e^{i(b+1)\theta_0} (\zeta^{j(b+1)} - 1), \quad \zeta = e^{2\pi i/d},$$

where $\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx > 0$ for $s > 0$. The column factors are all nonzero. A dependence among the remaining columns would give a polynomial $\sum_{l=1}^q b_l(Z^l - 1)$ of degree at most q , vanishing at all $q+1$ roots of unity, including 1. This polynomial is zero, so every $b_l = 0$.

Join the monomial coefficients to the given (c, t) by a straight path with real parameter $s \in [0, 1]$, keeping u fixed. On this compact coefficient path the leading negative ray term dominates all lower terms uniformly: for sufficiently large r the absolute value of any fixed coefficient derivative of the integrand is at most $Cr^L \exp[-r^d/(2d|u|)]$. This is integrable by the same gamma substitution; hence differentiation under every contour integral is valid. A derivative in c_b multiplies the integrand by $S^{b+1}/((b+1)u)$, and one in t multiplies it by $-S/u$. The unique reduction proved above, together with the vanishing of $\int_{\Gamma_j} e^{(\Phi-tS)/u} D_{u,t} P dS$, thus gives $\Pi'(s) = \Pi(s)B(s)$ with a continuous matrix $B(s)$. That integral vanishes because it is u times an exact derivative; both infinite endpoint values are zero and the two finite endpoint values cancel with the specified contour signs. The determinant identity

$$\frac{d}{ds} \det \Pi = \text{Tr}(\text{adj}(\Pi)\Pi B) = (\det \Pi) \text{Tr} B$$

holds also at singular matrices. Therefore $\det \Pi(s) = \det \Pi(0) \exp(\int_0^s \text{Tr} B(v) dv) \neq 0$. This proves invertibility without removing repeated-root parameters. Equation DT.6 proves it for the translated matrix as well.

Let G_N be any one of the original canonical arithmetic Gram matrices, and retain its index N , cochain degree k , and control form $W_N = A^* G_N + G_N A - k G_N$. The exact pullback along DT.2 is

$$G_N^{[a]} = C_a^* G_N C_a. \quad (\text{DT.7})$$

This is a transported specified form. No claim identifies it with a newly computed canonical theta form for a translated polynomial.

Proposition 3 (Control and determinant transport). *For the transported form and the original value of k ,*

$$W_N^{[a]} = A_a^* G_N^{[a]} + G_N^{[a]} A_a - k G_N^{[a]} = C_a^* (W_N + (a + \bar{a}) G_N) C_a, \quad (\text{DT.8})$$

$$\text{Tr}((G_N^{[a]})^{-1} W_N^{[a]}) = \text{Tr}(G_N^{-1} W_N) + 2q \text{Re } a. \quad (\text{DT.9})$$

Write $H_\Pi = \Pi^* \Pi$ and $B_N^{\text{per}} = G_N^{-1} H_\Pi$. Then

$$B_N^{\text{per},[a]} := (G_N^{[a]})^{-1} \Pi_a^* \Pi_a = |f_a|^2 C_a^{-1} B_N^{\text{per}} C_a. \quad (\text{DT.10})$$

Consequently, for the same (u, t) and the same original indices,

$$\frac{\det B_j^{\text{per},[a]}}{\det B_i^{\text{per},[a]}} = \frac{\det B_j^{\text{per}}}{\det B_i^{\text{per}}} = \frac{\det G_i}{\det G_j}. \quad (\text{DT.11})$$

Proof. Insert $C_a A_a C_a^{-1} = A + aI$ into the first definition of DT.8. Taking the inverse Gram and using invariance of trace under similarity gives DT.9. Equation DT.6 gives $\Pi_a^* \Pi_a = |f_a|^2 C_a^* H_\Pi C_a$, proving DT.10. Its determinant contains the factor $|f_a|^{2q}$; this same factor occurs at both indices and cancels in their ratio. The remaining period Gram also cancels, giving DT.11. Thus the original endpoint determinant cost is retained by this exact comparison. $\square$

4 The finite-field comparison with its scalar character

Let $R_0 \subset \mathbb{C}$ be the finitely generated coefficient ring of the original exponential construction, and adjoin a . For a specified map $R_0[a] \rightarrow \mathbb{F}_Q$ of characteristic $p > d$, and specified $u_0 \in \mathbb{F}_Q^\times$, $t_0 \in \mathbb{F}_Q$, reduce every displayed coefficient through that map. Put

$$f(X) = \frac{\Phi(X) - t_0 X}{u_0}, \quad \gamma_a = \frac{-\Phi(-a) - t_0 a}{u_0} \in \mathbb{F}_Q.$$

Use the Artin–Schreier rank-one sheaf convention in which the geometric-Frobenius trace at $x \in \mathbb{F}_{Q^n}$ is $\psi(\text{Tr}_{\mathbb{F}_{Q^n}/\mathbb{F}_p} f(x))$, for a fixed nontrivial additive character ψ valued in $\overline{\mathbb{Q}}_\ell$, $\ell \neq p$.

Proposition 4 (Translation and the constant character line). *For the isomorphism $\sigma_a : X \mapsto S = X + a$,*

$$\sigma_a^* \mathcal{L}_\psi((\Phi_a - t_0 S)/u_0) \simeq \mathcal{L}_\psi(f) \otimes \mathcal{L}_\psi(\gamma_a). \quad (\text{DT.12})$$

Hence compactly supported cohomology in every degree is identified with that of $\mathcal{L}_\psi(f)$, tensored with the displayed constant character line. Each geometric-Frobenius eigenvalue is multiplied by

$$\zeta_a = \psi(\text{Tr}_{\mathbb{F}_Q/\mathbb{F}_p} \gamma_a). \quad (\text{DT.13})$$

This is a root of unity, so it preserves each complex absolute value and hence the purity weights of this finite-field realization.

Proof. The potential identity in the period proof holds after the specified coefficient reduction. The tensor identity of Artin–Schreier character sheaves can be obtained directly from their covers: if $y_1^p - y_1 = f$ and $y_2^p - y_2 = \gamma_a$, then $(y_1 + y_2)^p - (y_1 + y_2) = f + \gamma_a$. The addition map identifies the contracted product of the two $\mathbb{F}_p$ -torsors with the third torsor. On the character lines, the addition action becomes the product of their characters, which gives DT.12. Pullback by the base-defined isomorphism σ_a preserves compact supports and commutes with Frobenius. Tensoring with a line pulled back from the base tensors the compact-support complex with that line: this can be checked on a cochain resolution before taking cohomology. Our stated trace convention makes its Frobenius scalar DT.13. The additive group of $\mathbb{F}_p$ has exponent p , so $\zeta_a^p = 1$, proving the absolute-value assertion. $\square$

These formulas compose with the exponential-family application of Deligne, *La conjecture de Weil II*, §§3.7.2–3.7.4, to the degree- d polynomial with nonzero leading coefficient. The source for that application, including the boundary chart and its hypotheses, is retained in the exponential comparison note; the present proof establishes the additional translation maps themselves.

5 Specialization and split support

The polynomial chain isomorphism T_a is defined before either complex or finite-field realization. Its coefficient map, the specialization $u = t = 0$, and the inclusion of the original fibre $a = 0$ induce maps under the same split-zero functor G . For a ring map $\varphi : B \rightarrow B'$, this functor acts by

$$G(\varphi)(\tau) = \tau, \quad G(\varphi)(b^\bullet) = \varphi(b)^\bullet.$$

Therefore a supported coefficient specialized to zero maps to $e = 0^\bullet$, and an absent coefficient maps to τ . The square with the amplitude maps $p_B, p_{B'}$ commutes, since both composites take τ to zero and $b^\bullet$ to $\varphi(b)$. On each already specified receiving module fibre, the coefficient chain relation specializes to the relation $\chi P = 0$ without changing its label. The functorial map thus retains both the supported kernel and external absence.

The translation formulas specify exactly how the two realizations respond to the same coefficient operation: DT.2 shifts the arithmetic sum operator, DT.9 changes its metric trace by $2q \text{Re } a$, DT.10 retains the analytic exponential scalar, and DT.13 supplies the finite-field character scalar. These are complete comparison maps. They do not identify a finite-field Frobenius with A , and they leave the original theta metric estimate as the same determinant quantity in DT.11. The original arithmetic construction remains the specified fibre $a = 0$ throughout.